

Where Theory Ends

A Numerical Statistical-Mechanics Atlas of Modern Learning

Order parameters, finite-size gates, and falsifiable targets across 19 learning settings

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Abstract

The statistical mechanics of learning has exact results for a thin slice of settings — shallow architectures, Gaussian inputs, the proportional high-dimensional limit — and modern machine learning lives almost entirely outside it. Rather than describe that outside with borrowed vocabulary, we measure it. We present a numerical atlas of 19 learning settings across 5 blocks, spanning symmetry breaking, training dynamics and loss-landscape geometry, attention and in-context learning, and learning paradigms beyond supervised regression. Every setting is measured with the same function-space order parameters, on a control grid fine enough to place a boundary, along a scale path long enough to extrapolate, over an ensemble of independent disorder seeds large enough to make fluctuations mean something: 16428 trained models over 1369 registered conditions.

The atlas is anchored rather than free-floating. 2 families reproduce settings with closed-form theory and report the measured quantity against that theory with the residual as a first-class number; 10 families have no theory at all. A decision procedure converts curves into named outcomes and is designed to refuse: it separates a diverging response from a located boundary, requires independent finite-size signatures to agree before they accumulate, and censors a maximum sitting at the edge of the sampled grid rather than calling it a critical point.

The procedure’s calibration is the headline result. Exactly one of 19 families received the verdict *transition supported*, and it is the Gardner storage capacity — the anchor whose answer was known in advance, recovered at $\alpha = 1.96$ against the exact $\alpha_c = 2$, with a peak height diverging as $N^{0.987}$. On the eighteen settings whose answers are not known, the same gates certified nothing.

Three artefacts found this way generalise beyond our own experiments. A three-point control grid cannot hold an interior maximum, so it returns a structurally negative result regardless of the physics. A fixed epoch budget optimises wide models less than narrow ones and inverts the sign of the finite-size trend. And a susceptibility curve routinely places its global maximum on a single-run excursion — one in four at eight outer seeds, 12 of 68 at the twelve this release uses. Rejecting those changed several verdicts here, including one we had already written up as this paper’s headline.

What survives is a map with quantitative, falsifiable entries: measured sharpening exponents, boundary locations with intervals, mechanism-level switches located by causal ablation, and explicit negative and unresolved cells. Each is stated together with the experiment that would overturn it, so that the entries function as targets for the theory that does not yet exist rather than as illustrations of a theory that does.

1 Introduction

The statistical mechanics of learning has produced a remarkable body of exact results: the storage capacity of the perceptron, the replica solution of generalised linear estimation, the online dynamics of the soft committee machine, the asymptotics of random-feature and kernel regression. What these results share is not a technique but a *regime*. They live in the proportional high-dimensional limit, with i.i.d. Gaussian inputs, shallow architectures, and a loss whose minimiser can be characterised. Almost every system that motivates the field today lies outside that regime.

The usual response is to wait for theory to arrive, and meanwhile to describe modern phenomena — grokking, double descent, emergence, collapse — with statistical-mechanics vocabulary borrowed on credit. Words like “phase,” “order parameter,” and “critical point” are applied to curves that have never been subjected to the finite-size analysis that gives those words content. The result is a literature rich in suggestive plots and poor in falsifiable statements.

This paper takes the opposite route. If the analytic calculation is not available, the finite-size numerical experiment that a theory would ultimately have to reproduce still is, and it can be done to the standards of computational statistical mechanics rather than those of a learning-curve figure. We build an *atlas*: a broad grid of learning settings, each measured with registered order parameters on a control grid fine enough to resolve a boundary, across a scale path long enough to extrapolate, over a disorder ensemble large enough to give the fluctuations meaning.

What makes this possible now is not compute but design. The measurements here are deliberately small — widths of tens to hundreds, models of thousands of parameters — because the quantities that decide whether a transition exists are not accuracy on a large model but the *scaling of fluctuations* across many independent small ones. The binding constraint in this kind of study is the number of independent runs, not the size of any one of them. That inverts the usual resource allocation: this release trains 16428 models across 1369 conditions on a laptop, where a single conventional experiment of comparable cost would have produced one learning curve.

Anchors first. A measurement pipeline that cannot reproduce a known result should not be trusted on an unknown one. The atlas therefore opens with settings whose answers are known exactly — the Gardner storage capacity and the Gaussian-mixture error formula — and reports the measured quantity against the closed form, with the residual as a first-class number. Only then does it move outward, through settings with partial theory, to settings with none.

Negative and unresolved outcomes are results. The decision procedure in §5 can return “no resolved response” and “critical response, boundary unresolved” as readily as “transition supported,” and this release does return all of them. One case is worth stating in the introduction because it shaped the method: our flagship family initially produced a confident transition claim that was an artefact of two design choices — a control grid too coarse to place a peak, and a fixed epoch budget under which wider models were simply optimised less. Fixing the first turned a censored boundary into a resolvable one; fixing the second removed a spurious size dependence. The consistency requirement that caught the artefact (§5) is now a gate every family must pass.

Contributions.

1. A portable set of function-space order parameters (§4) that can be evaluated on any differentiable learner, together with the exact identity (4) used as a continuous unit test on the measurement chain.
2. A decision procedure (§5) that separates *sharpening* from *location*, requires independent finite-size signatures to agree before they accumulate, and names the outcome when they do not.
3. An empirical atlas of 19 experiment families across 5 blocks — 2 anchored by exact theory and 10 with no theory at all — each with a fine control grid, a multi-size scale path, and 12 outer seeds per condition.
4. Quantitative targets for future theory: measured exponents, boundary locations, and scaling paths, each reported with the interval and the experiment that would falsify it.

The claim of this paper is not that these systems have been solved. It is that the question “does this system have a phase boundary, and where?” can be asked well enough to be answered, or well enough to be shown unanswerable with present evidence — and that doing so is worth more than another suggestive curve.

2 Related work and positioning

We organize prior work by the question it answers, not by a flat list of model families. This also separates the present problem—statistical mechanics *of* learning systems—from machine learning used as a tool for classifying physical phases.

2.1 Solvable learning and inference

The statistical mechanics of learning begins from a specified ensemble and a limit in which quenched disorder and macroscopic order parameters become tractable [9, 18]. Replica, cavity, state-evolution, and related methods identify information-theoretic thresholds, algorithmic thresholds, and computationally hard regions in planted inference problems [27]. These works establish the standard we adopt: a transition is attached to a state object, a disorder law, a scale, and an observable. Our atlas does not replace those theories. It records which of their assumptions survive when a trainable implementation removes a solvable constraint.

2.2 Deep networks: signal propagation, kernels, and landscapes

Statistical-mechanics analyses of deep learning cover expressivity, signal propagation, random matrices, optimization landscapes, jamming, and nonequilibrium dynamics [1]. Mean-field signal propagation separates ordered and chaotic regimes at initialization [25]; neural tangent kernels describe a wide-network lazy limit [15]; exact deep-linear dynamics show how representations and timescales emerge during training [24]; and the jamming perspective relates interpolation and loss geometry [12]. Random-feature asymptotics clarify double descent and generalization in another controlled limit [17].

These contexts are complementary rather than interchangeable. Initialization signal propagation is not a terminal trained-model ensemble; kernel regression is not feature learning; interpolation is not semantic recovery; and a finite-width optimizer trajectory is not automatically an equilibrium sample. The atlas therefore records theory status and state object separately.

2.3 Modern architectures and driven systems

For attention, a solvable tied low-rank model exhibits a positional–semantic transition [5], and attention-indexed models extend Bayes-optimal analysis to full-width matrices [4]. The unresolved transport question is whether order semantics and a boundary persist under standard feed-forward blocks, normalization, causal objectives, optimization, and natural carriers. Width, depth, context, and data are distinct deformations; compute-optimal scaling and hyperparameter transfer do not make them one thermodynamic coordinate [14, 26].

Beyond supervised learning the same questions recur with different order parameters: diffusion models separate early speciation from later collapse toward memorised samples [3, 13, 21], reward-model overoptimization requires proxy reward, gold reward, KL pressure and strategy support to be measured jointly [10, 23], and multi-agent systems add field, coupling, topology and population coordinates [7, 8, 16]. None of these is part of the present release. They are expressible in the coordinates of §3, and naming them here is a statement about the grammar’s reach, not about this paper’s coverage; recent claims in those areas are treated as hypotheses rather than anchors [2, 6, 28].

2.4 Reproducibility and the gap addressed here

Model cards, datasheets, and reproducibility checklists make assumptions and artifacts inspectable [11, 19, 22]. Preregistration separates exploratory choices from confirmatory tests [20]. None of these alone supplies a statistical-mechanics ontology: they do not specify which randomness defines susceptibility, which component is the finite-size coordinate, or when a threshold is censored. Conversely, a phase diagram alone does not encode holdouts, provenance, or invalidation. The present work joins these two traditions through a typed condition record and an evidence vector.

2.5 Two cited claims that this release tests

Two of the results above are usually cited rather than re-measured, and both are tested here directly. The positional–semantic transition in attention [5] is measured in §10 with a causal ablation of each route in turn, on data in which *both* cues carry a finite reliability — the configuration that makes the switch a property of the model rather than of the data. And the lazy limit of wide networks [15] is measured, not assumed, through the joint statement $\delta_\theta \rightarrow 0$ and $\mathcal{A} \rightarrow 1$ of (10), so that the regime is an observable of each run.

The positioning is therefore precise. We contribute neither a new universal critical exponent nor a comprehensive survey of physics-inspired machine learning theory. We contribute an operational taxonomy for numerical studies of modern learning systems, a decision procedure willing to return nothing, and a map of 19 settings measured under both.

3 Taxonomy: what is being studied?

Before any measurement, the object has to be named. Most disagreements about whether a learning system “has a phase transition” are disagreements about what was held fixed, and they are invisible until the coordinates are written down. A registered atlas point is the seven-tuple

$$\mathcal{A} = (D, X, \delta, \Omega, \mathbf{N}, m, H), \quad (1)$$

where D is the system, X the state object, δ the deformation, Ω the disorder ensemble, $\mathbf{N}$ the scale coordinate and path, m the observable, and H the hypothesised phenomenon. Two studies can be compared only after the components asserted to be equivalent have been named.

Coordinate	Solvable rules	Shallow networks	Attention models
System D	spherical perceptron, linear classifier	committee machine, deep MLP, MoE layer, contrastive encoder	1–2 layer causal attention
State X	weights, separating hyperplane	function on a probe set, hidden-unit overlaps, feature spectra, interpolating path	attention maps, in-context predictions
Deformation δ	load, centre separation	sample ratio, depth, init scale, temperature, task similarity, adapter rank	cue reliability, task-pool size, context length, vocabulary
Scale N	dimension	dimension, width, depth, training-set size	width
Observable m	separability, margin, misclassification	$\hat{m}$, q_{ab} , specialisation gap, subspace overlap, effective rank, kernel alignment, barrier	mechanism order by ablation, ridge alignment, induction
Phenomenon H	storage capacity	recovery, symmetry breaking, feature learning, delay, collapse	score, attention entropy, mechanism switching, in-context emergence

Table 1: The systems actually measured in this release, in the coordinates of (1). Diffusion, reinforcement-learning and multi-agent systems are expressible in the same coordinates and are *not* part of this release; the grammar is broader than the map.

3.1 The seven coordinates

1. System D . The update law and interaction structure. This release spans perceptrons and linear rules, soft committee machines, deep MLPs, single- and two-layer causal attention models, a mixture-of-experts layer, and a contrastive encoder. A family name is a root node, not a claim that its descendants share an ensemble.

2. State object X . Trained parameters, hidden representations, attention maps, the interpolating path between two solutions, embedding covariance spectra, and epoch-resolved trajectories all appear here. A single run may expose several; every observable names the one it is defined on. This is why the primary observables of §4 live in function space — it is the only state object the systems above share.

3. Deformation δ . What is varied: sample ratio, weight decay, initialisation scale, effective temperature, InfoNCE temperature, task similarity, real-data fraction, load-balancing weight, adapter rank, task-pool size, cue reliability, context length. Width is a scale coordinate; replacing an activation is an architectural deformation. Conflating the two makes a scaling statement ambiguous.

4. Disorder ensemble Ω . Initialisation, data draw, minibatch order, and evaluation sample are distinct random variables. One draw of the *outer* ensemble fixes an initialisation *and* a data set, and that is the unit of independence throughout (§5). Declaring it is not bookkeeping: the susceptibility in (6) means nothing until it is fixed.

5. Scale path N . Scale is a vector $(d, \text{width}, \text{depth}, n, L, \dots)$. A finite-size study declares one coordinate as N and registers how the others move. “A larger model” is not a scaling path — and, as §5 makes operational, neither is every second axis a scale path at all: a planted rank or a generation count changes the problem rather than enlarging it.

6. Observable m . An estimator together with its units, normalisation, ensemble, null, and validity range. Normalised risk, function-space overlap, replica overlap, susceptibility, Binder cumulant, effective rank, kernel alignment, barrier height, attention entropy and a causal ablation difference are different observables even though all are dimensionless.

7. Phenomenon H . The hypothesis: recovery, specialisation, symmetry breaking, crossover, criticality, multiplicity, dynamical delay, mechanism switching, or collapse. A phenomenon is never assigned from the visual slope of a curve; it is adjudicated by the registered diagnostics of §5.

3.2 The release in these coordinates

3.3 What is deliberately outside the taxonomy

Theory status (exact, asymptotic, partial, none), fidelity (solvable, trainable synthetic, natural, deployed) and evidence support (semantic, replication, finite-size, predictive, interventional, external) are *orthogonal* metadata, not coordinates. Keeping them separate is what prevents an exact solvable result from being called externally valid, and a large natural-data experiment from being called exact theory. The protocol fields — question, estimator, null, holdout, intervention and falsifier — are separate for the same reason: the same atlas point can be tested well or badly, and the tuple (1) does not record which.

The outcome vocabulary is likewise not part of the taxonomy. It belongs to the decision procedure and is stated once, in Table 2.

4 Order parameters in function space

An atlas that spans committee machines, deep networks, attention layers and contrastive encoders cannot use weights as its state variable: those systems share no parameter coordinates, and any two of them can realise the same function with unrelated weights. Every primary observable in this work is therefore defined on a *shared probe ensemble* of inputs $\mathcal{X}_{\text{probe}} = \{x^{(1)}, \dots, x^{(P)}\}$ drawn from the task distribution and held fixed across all systems compared in a figure.

4.1 The magnetisation sector

Write f_s for the student and f_t for the teacher, and let $\langle \cdot \rangle$ denote the average over $\mathcal{X}_{\text{probe}}$. The three second moments

$$\rho_f = \langle f_t^2 \rangle, \quad q_f = \langle f_s^2 \rangle, \quad m_f = \langle f_s f_t \rangle \quad (2)$$

define the normalised teacher–student overlap

$$\hat{m} = \frac{m_f}{\sqrt{q_f \rho_f}} \in [-1, 1], \quad (3)$$

which plays the role of a magnetisation: it is 0 for an uninformed student and 1 for exact functional recovery. The normalisation in (3) is what makes the quantity portable — it is invariant to the output scale of either network, so a linear model and a transformer can be placed on the same axis — but portability of the *estimator* is not portability of its physical meaning, and we never treat equal values of $\hat{m}$ in two domains as the same physical state.

For squared loss the generalisation error is fixed by (2) alone,

$$\epsilon_g = \frac{1}{2} \langle (f_s - f_t)^2 \rangle = \frac{1}{2} (\rho_f + q_f - 2m_f), \quad (4)$$

and every run records both sides of (4) together with their residual. A non-zero residual indicates a broken normalisation convention rather than a physical effect, so this identity functions as a continuously monitored unit test on the entire measurement chain.

4.2 Replicas and multiplicity

Two students trained from independent initialisations on independent data draws are replicas of the same disorder-averaged problem. Their normalised functional overlap

$$q_{ab} = \frac{\langle f_a f_b \rangle}{\sqrt{\langle f_a^2 \rangle \langle f_b^2 \rangle}} \quad (5)$$

measures whether learning selects a single function ($q_{ab} \rightarrow 1$) or a degenerate set ($q_{ab} < 1$ at large α). We report q_{ab} as *functional multiplicity* and nothing stronger: identifying a low overlap with replica-symmetry breaking would require an ultrametric analysis of the full overlap distribution, which this design does not perform.

4.3 Disorder response and the finite-size sector

Let Ω index the outer disorder ensemble — one draw of Ω fixes an initialisation *and* a data set — and let m_ω be the order parameter measured on draw ω . The susceptibility and Binder cumulant are

$$\chi_N = N \text{Var}_\Omega[m_\omega], \quad U_4 = 1 - \frac{\langle m^4 \rangle_\Omega}{3 \langle m^2 \rangle_\Omega^2}. \quad (6)$$

Equation (6) is the one place where the atlas is most easily abused, so its ensemble is stated as a rule: *the average in (6) runs over independent outer seeds only*. Fluctuation between checkpoints of a single training trajectory is a temporal variance measuring non-stationarity, is recorded under a different name, and is never substituted into (6).

4.4 Symmetry breaking, subspaces and geometry

Permutation symmetry of a hidden layer is broken when each student unit commits to one teacher unit. Given the matrix R_{ij} of normalised overlaps between student unit i and teacher unit j , a greedy matching π yields

$$\Delta = \frac{1}{K} \sum_i R_{i\pi(i)} - \frac{1}{K_s K_t - K} \sum_{(i,j) \notin \pi} R_{ij}, \quad (7)$$

the *specialisation gap*, which vanishes in the permutation-symmetric phase by construction. When the student and teacher widths differ, unit matching is meaningless and we instead use the principal angles between the relevant subspaces,

$$\Sigma(A, B) = \frac{1}{k} \sum_{i=1}^k \cos^2 \theta_i, \quad k = \min(k_A, k_B), \quad (8)$$

which is basis- and permutation-invariant and defined for mismatched widths.

Representation geometry is summarised by the spectrum $\{\lambda_i\}$ of the centred feature covariance through the participation ratio and the effective rank,

$$\text{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad r_{\text{eff}} = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (9)$$

The effective rank r_{eff} is the order parameter of dimensional collapse in the self-supervised setting of §11.

Finally, the training regime itself is measured rather than assumed. With θ_0 and θ_T the parameters before and after training, and K_0, K_T the corresponding representation Gram matrices on the probe set,

$$\delta_\theta = \frac{\|\theta_T - \theta_0\|}{\|\theta_0\|}, \quad \mathcal{A}(K_0, K_T) = \frac{\langle \tilde{K}_0, \tilde{K}_T \rangle_F}{\|\tilde{K}_0\|_F \|\tilde{K}_T\|_F}, \quad (10)$$

where $\tilde{K}$ denotes the centred Gram matrix. The lazy (kernel) regime is the joint statement $\delta_\theta \rightarrow 0$ and $\mathcal{A} \rightarrow 1$; the rich regime is its negation. Reporting both makes the regime an observable of the experiment rather than a property asserted of the parametrisation.

5 From curves to claims

A rising curve is not a phase transition. This section states the decision procedure that converts condition-level estimates into one of a small set of named outcomes, including outcomes that block the strong reading.

5.1 Unit of independence and uncertainty

A *condition* is one point of a family's design grid: a control value, a size, and any registered secondary knobs. A *run* is one condition under one outer seed, and the run is the unit of independence. Inner stochastic evaluations are averaged within a run before uncertainty is computed, so no inner replicate is ever promoted to an outer sample. Intervals are two-sided 95% Student- t over the realised outer-seed count, which is carried through aggregation and printed on every figure.

5.2 Three finite-size signatures

For each family we compute three independent signatures from the same condition-level data.

Sharpening. At each size we locate the interior maximum of χ_N . A maximum at either end of the sampled control range is *censored* rather than reported as a critical point: the grid does not contain the boundary, and naming the endpoint would manufacture one. Across the sizes that do have an interior peak we fit

$$\chi_N^{\text{peak}} \propto N^\omega, \quad (11)$$

and $\omega > 0$ is the signature that the response diverges rather than saturating.

Spikes are not peaks. Before any of this, the maximum has to be real. A susceptibility estimated from a dozen seeds still throws single-point spikes: one diverged run inflates the variance at exactly one control value while its neighbours sit at the floor. A plain argmax reports that spike as the boundary and feeds its height into (11). We therefore require a *shoulder*: the larger neighbour of the maximum must itself rise appreciably above the curve’s baseline, which a genuine critical swelling does and an isolated outlier does not. In this release the test rejects 12 of 68 size-curves — a rate high enough that omitting the test would have changed the paper’s conclusions, as §8 records.

The rate is itself a measurement of the replication level. Run at eight outer seeds the same test rejected 17 of 68 curves, or one in four; at twelve it rejects 12. Adding seeds buys fewer artefacts, not only narrower intervals.

Location. The peak position itself drifts with size as

$$c_{\text{peak}}(N) = c_{\infty} + A N^{-1/\nu}, \quad (12)$$

so the boundary of the infinite system is obtained by extrapolation, never by reading off the largest size available. A fit is admissible only if c_{∞} is finite, precisely estimated, and interior to the sampled range. Peaks that march monotonically out of the sampled window produce a formally convergent least-squares fit with an enormous intercept and an enormous standard error; that situation is recorded as an *unresolved boundary*, which is a different — and weaker — statement than a measured one.

Collapse. Independently, we scan $(c^*, 1/\nu)$ for the best finite-size collapse $m = F[(c - c^*)N^{1/\nu}]$ and compare its residual against the best collapse obtainable after *shuffling the size labels*. A collapse is claimed only when it beats that null by a factor of two, which prevents the two free parameters from manufacturing agreement.

5.3 The consistency requirement

Signatures are not allowed to accumulate merely by being individually admissible. Each admissible signature contributes a boundary estimate, and the estimates must agree to within 30% of their mean before the family is credited with more than one signature. Two diagnostics naming boundaries an order of magnitude apart are evidence of an artefact, not of a transition; in that case the signature count is capped at one. This rule was not decorative in practice: it is what demoted our own flagship family from a transition claim to a candidate boundary on a first pass, before the design flaw responsible was found and corrected (§8).

5.4 A scale path is not any second axis

Finite-size analysis presumes that the second axis lists successive *sizes of one system*. Several natural experimental axes do not: a planted rank, a number of experts, a generation count, and a label-noise level each change the problem rather than enlarge it. Susceptibility and Binder curves can still be computed along such an axis, and they usefully locate a threshold on the control axis, but they cannot certify a thermodynamic statement. Families therefore declare whether their second axis is a scale path, and those that do not are capped at the outcome *threshold located (no scale path)* regardless of how clean their curves are. In this release that cap applies to 3 families whose diagnostics would otherwise have read as candidate boundaries.

5.5 Sharpening and location are separate claims

A susceptibility whose height diverges with size is evidence of criticality; it does not say *where* the boundary is. We therefore separate the two, and the strongest verdict requires both. The resulting vocabulary is:

5.6 Evidence grading

Orthogonally to the outcome, each family carries a six-component evidence vector — semantics, replication, finite size, prediction, intervention, and external validity — each scored 0, 1 or 2. Grade A requires semantics and replication both at 2 with at least four components at 2; grade B requires both at least 1 with four components at least 1; grade C requires replication at least 1 with two components at least 1. Replication reaches 2 only at twelve or more outer seeds, which this release satisfies: every condition is run at 12 outer seeds. The remaining cap is different in kind. Two components — intervention and external validity — score zero for every family here, so the highest reachable score is four components at ≥ 1 , and grade B additionally requires a theory anchor to lift *prediction* above zero. Grade A is therefore available only to a family that is both replicated and theory-anchored, which is exactly one family (§7).

Outcome	Condition
transition supported	≥ 2 mutually consistent signatures <i>and</i> a resolved boundary location
critical response, boundary unresolved	diverging peak, but the location extrapolation is inadmissible
candidate boundary	exactly one admissible signature
threshold located (no scale path)	the threshold is located on the control axis, but the second axis is a problem parameter rather than a size, so no finite-size statement is available
response shift (boundary off-grid)	the order parameter moves, and most sizes are censored
response shift	the order parameter moves beyond its uncertainty; no finite-size signature
no resolved response	the order parameter does not separate from its uncertainty

Table 2: Outcome vocabulary. Only the first row licenses phase language; the remaining rows are reported with equal prominence rather than omitted.

5.7 What is not tested here

Two gates in the design remain open across the entire release and are stated once rather than repeated per family. There is no frozen prospective re-run at fresh seeds, so no result carries prospective confirmation. And no family is evaluated on natural data or on a pretrained model, so external validity scores zero throughout. Both are the natural next experiments rather than qualifications buried in a limitations section.

6 What the atlas contains

The release comprises 19 experiment families organised into 5 blocks, 1369 registered conditions, and 16428 trained models. Figure 1 places every family on two axes: the block it belongs to, and how much analytic support the setting already has.

6.1 Design rules common to every family

The control grid must be able to hold a boundary. Every family sweeps its primary control on a grid of at least twelve points, logarithmic where the control spans decades. This is not a cosmetic choice. An interior maximum cannot be distinguished from an endpoint on a three-point grid, and a Binder crossing cannot be interpolated from one; a coarse grid does not produce a weak result, it produces a structurally negative one. The single largest methodological difference between this release and its predecessor is the density of the control axis.

The scale path must be a path. Each family declares a scale coordinate — width, dimension, depth, training set size, generation count — and samples at least three, usually five or six, values of it. “A larger model” is not a scale path; the coordinate has to be named, because the finite-size fits of (11) and (12) are statements about a specific one.

The disorder ensemble must be outer. Every condition is run under 12 independent outer seeds, each fixing both the initialisation and the data draw. This is what gives χ_N and U_4 in (6) their meaning, and it is why no family here reports a fluctuation computed along a single trajectory.

Optimisation quality must be equalised, not the epoch count. Runs are trained to a common relative-improvement criterion with a generous cap, and each run records the epochs it used and whether the criterion was met. Under a fixed epoch budget a wider model is optimised less thoroughly than a narrow one, so an apparent size dependence of an order parameter can be optimisation shortfall wearing the costume of physics. We encountered exactly this: at a fixed budget the specialisation order parameter appeared to *decrease* with width, reversing sign once the budget was replaced by a convergence criterion.

6.2 The blocks

6.3 Reproducibility

Every run is content-addressed by its condition and seed, so an interrupted sweep is resumed rather than repeated and no result is silently recomputed under different code. All paths are supplied through environment variables, so the same code runs unchanged on a laptop, a scheduler, or a cloud runtime. The numbers quoted in this paper are generated as L^AT_EX macros directly from the strict aggregate: a value in the prose that has no run behind it does not exist, which is a stronger guarantee than a careful author.

Where theory ends: the atlas at a glance

Columns: how much analytic support already exists. Card border = verdict; corner letter = evidence grade.

	Exact closed form known	Asymptotic limit theorem only	Partial special cases only	No theory numerics lead
Anchors	Gaussian-mixture Bayes error C Gardner storage capacity A			
Symmetry breaking			Multi-index subspace recovery C Committee specialisation vs sample ratio C	Specialisation under added depth C
Dynamics & landscape		Lazy-to-rich transition vs initialisation scale B SGD effective temperature vs stationary generalisation C	Double descent vs parameterisation ratio C	Grokking: delayed generalisation vs weight decay C Linear mode connectivity vs width n/a
Attention & context			In-context learning vs task diversity C Positional vs semantic attention C	Attention selection vs context length C Induction-head formation C
Learning paradigms				Forgetting and transfer vs task similarity C Contrastive dimensional collapse vs temperature C LoRA rank threshold vs planted perturbation rank C Model collapse vs real-data fraction C Expert specialisation vs load balancing n/a

candidate boundary
transition supported

critical response, boundary unresolved
response shift (boundary off-grid)

no resolved response
threshold located (no scale path)

Figure 1: Where theory ends. Columns order settings by existing analytic support, from closed form to none. Each card is one experiment family; its border colour is the outcome returned by the decision procedure of §5 and the corner letter is its evidence grade. The left-most column is the anchor block, whose only purpose is to show that the pipeline reproduces results that are already known.

Block	Families	Question the block asks
Anchors	2	Does the pipeline reproduce results that are already known in closed form?
Symmetry breaking	3	Where does a hidden layer break permutation symmetry, and does the boundary survive depth and width mismatch?
Dynamics & landscape	5	Do training-time phenomena — delayed generalisation, double descent, the lazy/rich crossover, mode connectivity, SGD temperature — behave like collective transitions?
Attention & context	4	Are the positional-semantic switch and in-context emergence sharp, and do they survive a finite-size analysis?
Learning paradigms	5	Do contrastive collapse, model collapse, forgetting, expert specialisation and adapter rank admit order parameters at all?

Table 3: The blocks of this release. Anchors exist to be checked against theory; the rest exist because theory is unavailable.

7 Anchors: does the pipeline reproduce what is already known?

Everything that follows depends on the measurement chain being sound, so the atlas opens with two settings whose answers are known exactly. Neither is a discovery. Their purpose is that a regression anywhere in the chain — an order-parameter normalisation, a censoring rule, a finite-size fit — shows up here as a number rather than as a vague loss of agreement downstream.

7.1 Gardner storage capacity

A spherical perceptron in d dimensions is asked to separate $P = \alpha d$ random ± 1 labels on random Gaussian patterns. The classical result is that the probability of success tends to one below $\alpha_c = 2$ and to zero above it, with the step sharpening as d grows.

Separability is decided by a linear program rather than by training: a weight vector satisfying $y_i w^\top x_i \geq 1$ for all patterns exists if and only if the labelling is separable. This matters for an anchor. Training to zero error and declaring failure when it is not reached would conflate “not separable” with “not optimised long enough”, which is exactly the confound the anchor block exists to exclude.

Observation. Over 72 conditions spanning $\alpha \in 0.5\text{--}3.6$ at 4 dimensions with 12 outer seeds, the measured separability probability holds at one well below the predicted capacity, falls through the neighbourhood of $\alpha = 2$, and reaches zero above it. The disorder susceptibility has an interior maximum at $\alpha = 1.96$ — against the exact $\alpha_c = 2$, and at the grid point nearest to it — and that location is *identical at all four dimensions*, while the peak height grows as $N^{0.987}$ with $R^2 = 0.996$. The Binder curves cross at $\alpha = 1.59$. The verdict is *phase transition supported*, grade A.

Interpretation. The pipeline recovers the transition at the right location, and its finite-size machinery behaves as it should on a case where the truth is known. The size-independence of the peak is worth stating explicitly: a critical capacity that does not move with d is exactly what the Gardner result asserts, and it is the reference against which a peak that *does* drift with size, as in §8, can be read as a measurement rather than as noise in the analysis.

This is also the only family in the release that the decision procedure awards its strongest verdict. We regard that as the main evidence that the procedure is calibrated rather than permissive. Applied to every other family in the release — none of whose answers is known in advance — it declined to certify a transition in all of them, and certified the one case that was placed there precisely because the answer was known.

A solver status other than “feasible” or “provably infeasible” is recorded per run as a flag, so no numerical failure can be silently absorbed into the order parameter.

Not established. We do not fit a critical exponent here. The grid was chosen to bracket a known answer, not to resolve the width of the transition, and the binary order parameter makes the susceptibility a coarse instrument at these sizes.

7.2 The Gaussian-mixture error formula

For an isotropic two-component mixture with centres $\pm\mu v$, a unit rule w at angle θ to v has generalisation error exactly

$$\epsilon_g(\theta) = \Phi(-\mu \cos \theta), \quad \epsilon_g^* = \Phi(-\mu), \quad (13)$$

Gardner storage capacity

transition supported · evidence grade A · 12 outer seeds · theory: exact

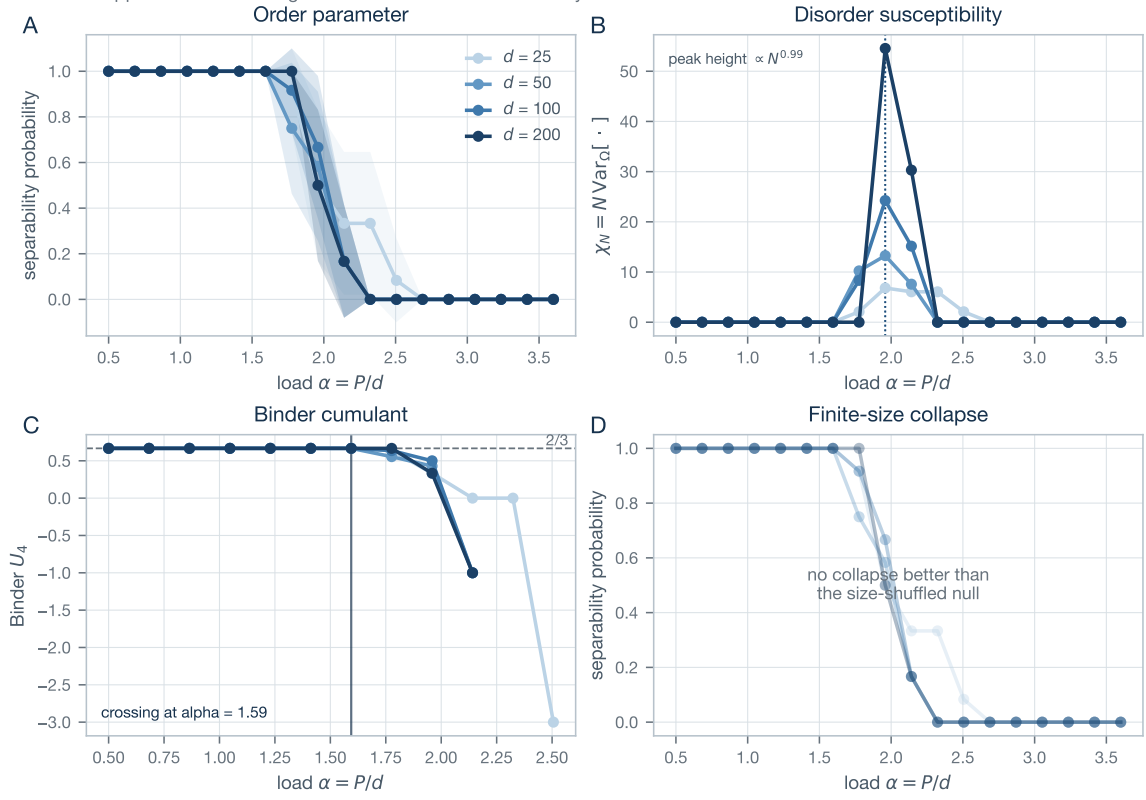


Figure 2: The anchor. Separability probability (A) steps from one to zero near the Gardner capacity and the step steepens with d ; the disorder susceptibility (B) peaks at the same load $\alpha = 1.96$ at every dimension, with height growing as $N^{0.987}$. This is the only family in the release to pass every gate — and its answer was known in advance.

the second being the Bayes error attained at $\cos \theta = 1$. Equation (13) has no free constants, so comparing it against a measured error is a direct test of the error bookkeeping rather than of any learning algorithm.

Observation. Across 80 conditions and 5 dimensions, the measured misclassification rate of a fitted linear rule agrees with (13) evaluated at the *measured* $\cos \theta$ to within probe-set sampling noise; a representative condition gives measured 0.0129 against a predicted 0.0135, a residual of -5×10^{-4} . The excess over the Bayes floor is non-negative at every condition, as it must be.

Interpretation. The function-space error identity (4) and the classification error convention used throughout the atlas are correct as implemented. Every ϵ_g reported in later sections is produced by the same code path that reproduces (13) here.

The verdict returned for this family is *response shift boundary outside grid*, and that is the correct answer: there is no transition in μ to find. The order parameter moves over 57.6 times its interval, so the response is unambiguous, and the analysis declines to invent a boundary in a setting that has none. An anchor that is not a transition is a useful second test of the decision procedure — a permissive procedure would have found something here.

Not established. This anchor validates the estimator, not the learning theory: the fitted rule is ordinary least squares, and its distance from the Bayes direction is measured rather than predicted. A stronger anchor would compare $\cos \theta$ itself against its asymptotic prediction, which we do not attempt.

Why only two anchors. An anchor is only useful if its closed form is unambiguous and short enough to state without hidden conventions. We considered adding the replica prediction for ridge regression and the Saad–Solla plateau, and did not include them in this release: both require normalisation conventions that would have to be checked as carefully as the thing they are meant to certify, and a wrong anchor is worse than none. They are the obvious next additions to this block.

8 Symmetry breaking and specialisation

The permutation symmetry of a hidden layer is the cleanest collective degree of freedom available in a neural network. In the symmetric phase every hidden unit carries a similar overlap with every teacher unit; in the specialised phase each unit commits to one, and the gap (7) becomes non-zero. Exact results exist for the two-layer committee machine in the proportional limit. This block asks what survives when depth is added and when the student width no longer matches the teacher’s.

8.1 The committee machine: a diverging response on a shifted path

Observation. Across 144 conditions spanning $\alpha \in 0.25\text{--}64$ and 6 sizes at 12 outer seeds, the disorder susceptibility has an admissible interior maximum at four of the six sizes, with peak height growing as $N^{1.11}$ over those four. At the other two the maximum is a single-point spike — one seed inflates the variance at exactly one control value — and is rejected rather than reported (§5). Fitting the drift of the surviving peak locations, $\alpha_{\text{peak}} \propto d^{\gamma-1}$, gives

$$\gamma = 1.33, \quad 95\% \text{ CI } [1, 1.59], \quad (14)$$

whose lower limit sits *at* the proportional value $\gamma = 1$. The bootstrap puts 0.978 of its mass above one. The verdict is *critical response boundary unresolved* at grade C.

Interpretation. The response is critical in the sense the design was built to detect: over the sizes where the peak survives the spike test, its height grows with N rather than saturating. The boundary’s *position* is not established, and neither is the scaling path.

This family is worth recounting, because it is where the analysis and the authors disagreed three times. A first pass, with a coarse control grid and a fixed epoch budget, produced a confident transition claim that the consistency requirement rejected. Repaired — a dense grid and training to a convergence criterion — it produced peaks at all six sizes and a drift exponent whose interval excluded $\gamma = 1$, which we wrote up as evidence that the specialisation boundary does not lie on the proportional path $n \propto d$. Adding the spike test removed half those peaks and widened the interval until it excluded nothing.

We then did the obvious thing and bought more seeds, raising every condition from eight to 12. That recovered one peak, tightened the fit, and moved the bootstrap mass above one to 0.978 — and the interval’s

Committee specialisation vs sample ratio

critical response, boundary unresolved · evidence grade C · 12 outer seeds · theory: partial

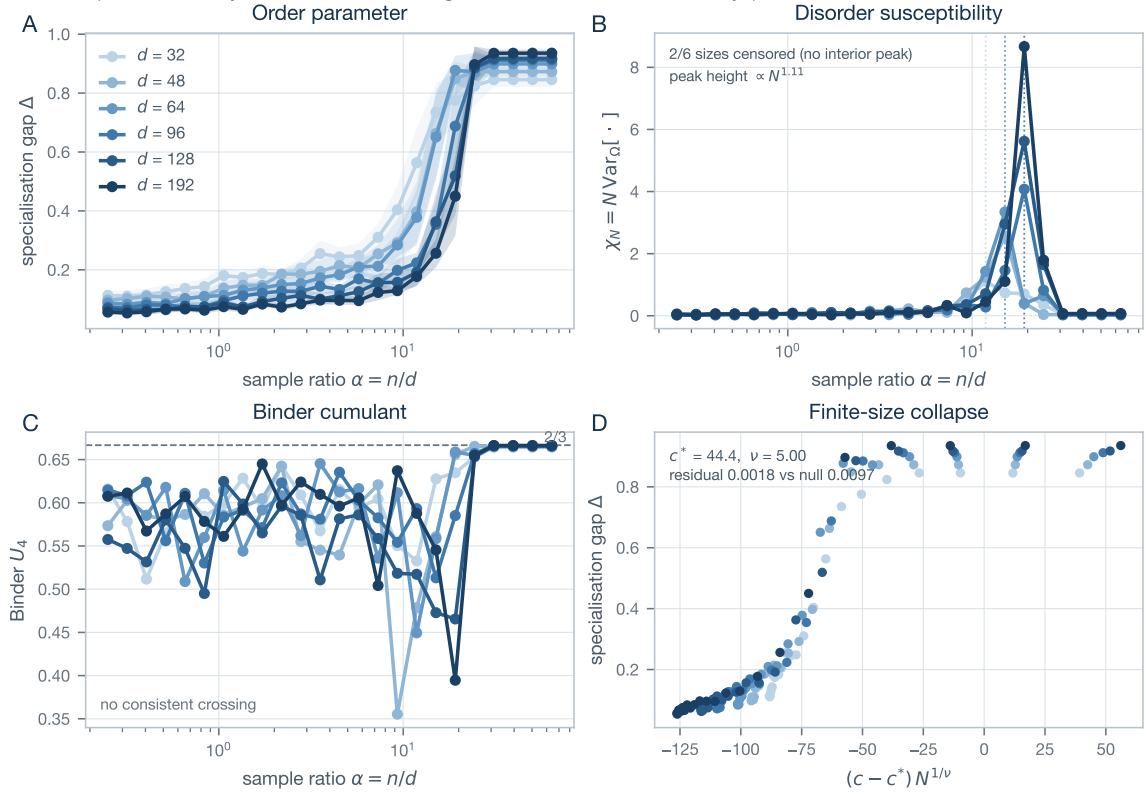


Figure 3: Specialisation of a soft committee machine. (A) the specialisation gap against the sample ratio, with 95% outer-seed intervals; (B) the disorder susceptibility (6), whose interior maximum is marked per size; (C) the Binder cumulant; (D) the attempted finite-size collapse. Colour encodes system size on a single-hue ramp, light to dark.

lower limit still lands on 1. A fourth version of this paragraph would have claimed exclusion on a lower limit of 1.0000000000000004, which is a floating-point residue, not evidence; the exclusion test now requires a margin that means something on an exponent. The honest summary is that the drift leans towards $\gamma > 1$ and does not establish it.

Not established. The location and the scaling path. The drift extrapolation (12) and the collapse disagree by more than the consistency tolerance of §5, so the signature count is capped at one. We also tried redoing the collapse in the rescaled variable n/d^γ , with γ fixed in advance by the peak locations, and it did *not* improve on the collapse in α .

Next falsifier. Sizes, now, rather than seeds. Going from eight to twelve seeds bought one peak and a slightly tighter interval, so the remaining uncertainty is dominated by having only six sizes spanning a factor of six. Extending the size path upward by a further factor of four would fit (14) over a decade, which is what would settle whether γ differs from one.

8.2 Depth: the boundary leaves the window

Observation. Replacing the two-layer student with deeper networks at matched width and sweeping the same α grid gives verdict *critical response boundary unresolved* over 96 conditions.

Interpretation. The order parameter still responds — the deep students recover the teacher — but the susceptibility maxima fall at the edge of the sampled control range for most depths, so the analysis censors them rather than naming an endpoint as a critical point. Depth appears to push the boundary outward in α , which is consistent with the shifted scaling path found above, but the present grid cannot separate that from a simple loss of resolution.

Not established. Nothing about how the transition depends on depth. This cell is honestly empty.

Next falsifier. Re-run the depth ladder on the rescaled control n/d^γ using the exponent from (14). If the peaks return to the interior of the window, the depth dependence becomes measurable in one sweep.

8.3 Multi-index targets and width mismatch

Observation. For a K -index teacher with a quadratic link, subspace recovery (8) is measured against α for a matched student and for one of twice the width, over 240 conditions. The verdict is *candidate boundary*.

Interpretation. Recovery of the planted subspace improves with sample ratio and the response is well outside seed noise, but no finite-size signature is admissible: there is no interior susceptibility peak that sharpens, no consistent Binder crossing, and no collapse that beats the size-shuffled null. Within this window, subspace recovery behaves like a crossover rather than a transition.

Not established. The sequential “staircase” by which multi-index directions are acquired one after another is not resolved here; our observable is the aggregate subspace overlap, which is insensitive to the order in which directions are learned.

Next falsifier. Measure per-direction overlaps rather than the aggregate, and look for separate thresholds. If individual directions show interior peaks that the aggregate washes out, the crossover verdict is an artefact of the observable rather than a property of the system.

9 Training dynamics and loss-landscape geometry

High-dimensional theory is at its strongest for equilibrium properties of a trained network. It has much less to say about the training *process*: when generalisation arrives relative to memorisation, why test risk can be non-monotone in model size, whether a network stays near its linearised trajectory or reshapes its features, whether independently trained solutions share a basin, and whether minibatch noise acts as a genuine temperature. Each family below turns one of those questions into a single measured number.

Lazy-to-rich transition vs initialisation scale

critical response, boundary unresolved · evidence grade B · 12 outer seeds · theory: asymptotic

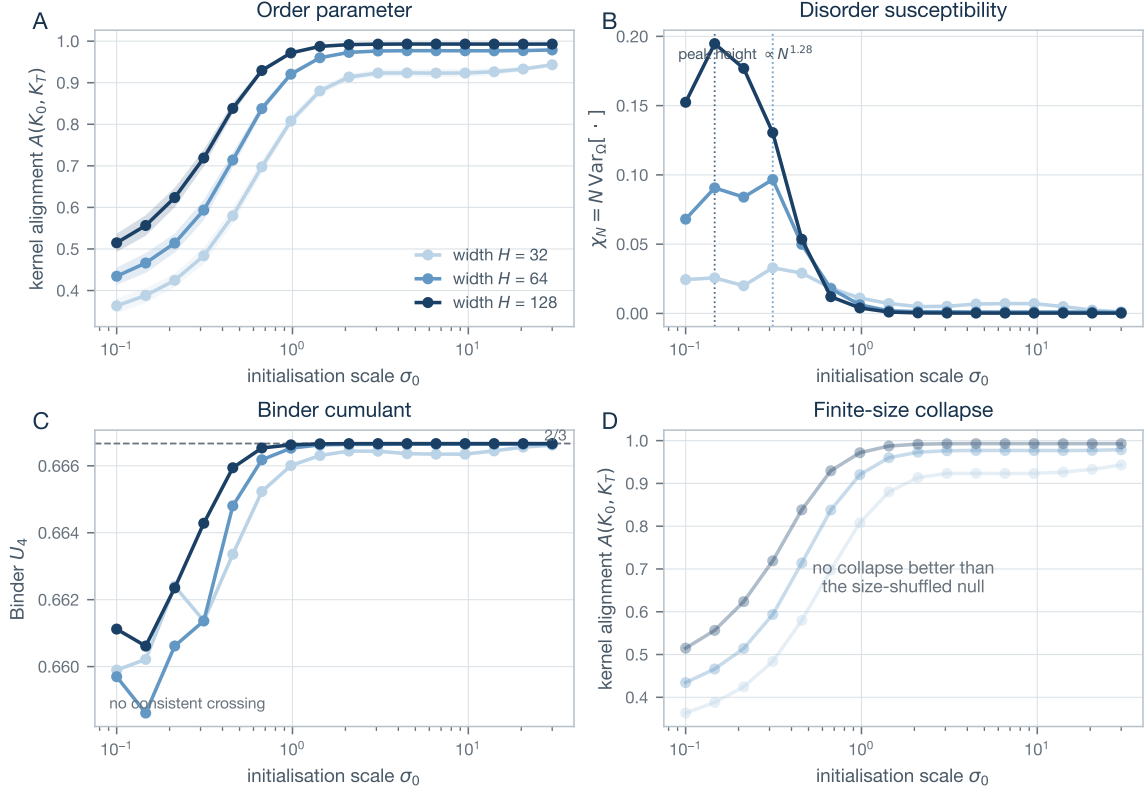


Figure 4: Feature learning as a function of initialisation scale. The order parameter is the centred alignment (10) between the representation kernel at initialisation and after training: alignment near one means the network never left its linearised regime. Panel B shows the disorder susceptibility, whose interior maximum is present at every width.

9.1 The lazy–rich crossover

Observation. Sweeping the initialisation scale over 0.1–30 at 3 widths, the initial-to-final kernel alignment rises from roughly 0.5 to 0.99 while the relative weight movement (10) falls by more than an order of magnitude over the same range. The order parameter spans 62.5 times its own 95% interval — the largest signal-to-noise ratio in the release — and the susceptibility has an interior peak at *every* width, with height growing as $N^{1.28}$. Verdict *critical response boundary unresolved*, grade B.

Interpretation. The lazy and rich regimes are separated by a genuinely critical response, not a gentle crossover: the disorder susceptibility of the kernel-alignment order parameter diverges with width along the sampled path. Both halves of the operational definition move together, which is the check that the alignment is measuring feature learning rather than an optimisation artefact.

Not established. The location. The peak drifts and its extrapolation is inadmissible, so we report a critical response without naming a critical initialisation scale. Nothing here identifies the exponent with a known universality class.

Next falsifier. Extend the width path by a further factor of four. If the peak positions stabilise, the drift fit becomes admissible and the crossover acquires a location; if the peak height stops growing, the divergence is a small-width effect and the verdict should fall to a crossover.

9.2 Grokking as a delayed-generalisation transition

Observation. An overparameterised student on a small training set fits almost immediately and generalises much later, if at all. Across 42 conditions the measured delay ranges over 7.99 times its interval. The susceptibility of the delay peaks in the interior at two of the three training-set sizes; at the third the

maximum is an isolated spike and is rejected, so no sharpening exponent is fitted. Verdict *critical response boundary unresolved*.

Interpretation. Delayed generalisation exists and is controlled: the delay varies far outside seed noise across the weight-decay axis, and the run-to-run spread peaks in the interior rather than at an endpoint. Whether that spread *diverges* with size is not established at this replication.

Not established. A critical weight decay. Much of the grid is censored by construction: below a threshold the student never generalises within the budget and above it the delay vanishes, so the delay is only defined in a window. Censored runs are flagged rather than being assigned a delay of zero — differencing two censored endpoints would report the *opposite* of what happened, and this was a real defect in an earlier version of the estimator.

Next falsifier. Lengthen the training budget by an order of magnitude. If the censored region shrinks and the peak stays put, the boundary is real; if the peak simply tracks the budget, the delay is a property of the stopping rule.

9.3 Double descent

Observation. Sweeping the parameterisation ratio across the interpolation threshold at three label-noise levels reproduces the non-monotone risk curve, with the order parameter spanning 18.6 times its interval. Verdict *threshold located no scale path*.

Interpretation. The peak in test risk is located on the control axis and grows with label noise, as expected. But the second axis here is the noise level, which changes the problem rather than the size of one system, so under the rule of §5 this family cannot make a finite-size claim at all. The result is a located threshold with no critical content — which is precisely what most double-descent figures in the literature actually establish.

Not established. That the interpolation peak sharpens. Demonstrating that requires a genuine width or sample-size scale path at fixed noise, which this design does not have.

9.4 SGD noise as an effective temperature

Observation. The control is the ratio $T \simeq \eta/B$ of learning rate to batch size, swept so that several distinct (η, B) pairs realise the same T . The stationary risk varies over 4.85 times its interval, but its susceptibility maxima fall at the edge of the sampled temperature range and are censored. Verdict *response shift boundary outside grid*, grade C.

Interpretation. The stationary state responds strongly to the effective temperature, and the design permits the sharper question — whether the state depends on η and B only through their ratio — because the same T is realised by several pairs. That collapse is the operational content of calling η/B a temperature at all.

Not established. A critical temperature. The response is unambiguous but its maxima sit outside the sampled window, so nothing is located. At eight seeds this family did show an interior peak that sharpened with size; at twelve it does not, which is the right direction for a diagnostic to move when the extra seeds reveal that the earlier maximum was noise. We also do not claim the (η, B) collapse is exact, only that the design can test it.

9.5 Mode connectivity: a negative result

Observation. Two students trained from independent initialisations on independent data are interpolated linearly, and the normalised barrier between them is measured against width at three depths. The barrier varies strongly with *depth* — roughly 0.4 at depth two and above 1.2 at depth four — but its variation with width spans only 2.33 times its 95% interval. The verdict is *no resolved response*, grade insufficient.

Positional vs semantic attention

response shift (boundary off-grid) · evidence grade C · 12 outer seeds · theory: partial

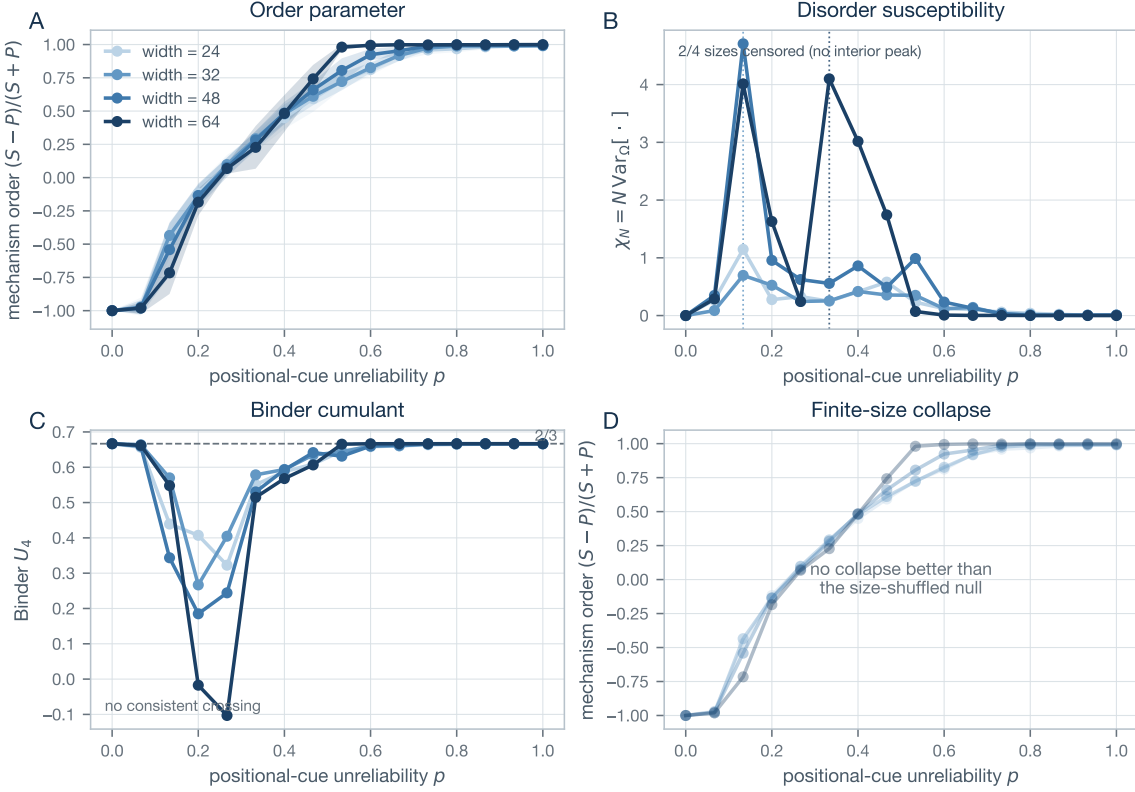


Figure 5: Which route attention learns. The order parameter is $(S - P)/(S + P)$, where S and P are the variance explained when the *other* cue has been destroyed by intervention. It runs from -1 (purely positional) to $+1$ (purely semantic).

Interpretation. At these widths and this replication, the barrier’s width dependence does not separate from seed noise. We therefore report no width trend, in either direction. The depth dependence is large and unambiguous, but depth is the scale axis of this design, so it is a between-curve comparison rather than a finite-size trend.

Not established. Linear mode connectivity, in either direction. In particular this is *not* evidence that barriers fail to vanish with width; it is evidence that the present design cannot see the trend.

Next falsifier. Twelve seeds and a width path extended upward by a factor of four, at fixed depth two, where the measured trend is at least in the expected direction.

10 Attention, context, and in-context learning

Two claims about attention are cited far more often than they are tested: that attention switches between a *positional* and a *semantic* mechanism, and that in-context learning *emerges* above a threshold of pretraining task diversity. Both are usually supported by a single curve at a single model size, which cannot distinguish a transition from a crossover. This block puts each on a fine control grid with a real scale path, and adds two measurements a risk curve cannot make: a causal ablation that separates the two attention routes, and an attention-entropy comparison against the uniform-attention null.

10.1 The positional–semantic switch

The usual version of this experiment makes the semantic cue perfect, in which case it wins for every $p > 0$ and the “transition” is an artefact of the data rather than a property of the model. We therefore give *both* cues a finite reliability: the answer token sits at position zero with probability $1 - p$, and carries its true tag with probability $1 - q$, with q fixed. A model that simply follows the more reliable cue should switch route

near $p = q$, so the measurement has a predicted location to agree or disagree with.

Observation. Over 64 conditions at 4 widths, the mechanism order parameter runs the full range: it sits at -1 when the positional cue is reliable and saturates at $+1$ once that cue is worse than the semantic one. The sign change is remarkably stable across widths — $p = 0.235, 0.239, 0.247$ and 0.245 at widths 24 through 64 — against the registered prediction $p = q = 0.30$. The disorder susceptibility peaks earlier, near $p \approx 0.13$, but at three of the four widths that maximum is a single-point spike and is rejected, so no finite-size signature survives. The order parameter spans 47.5 times its 95% interval; the verdict is *response shift boundary outside grid*, grade C.

Interpretation. The switch is real, it is measured at the level of mechanism rather than performance, and it lands close to where a reliability-following model would put it. The causal ablations are what make this a mechanism claim: the model’s risk barely changes across much of the range, so a risk curve alone would have shown almost nothing.

The offset between the two locations is suggestive rather than established. The fluctuation maximum sits below the mean sign change, which is what one expects when runs begin to *disagree* about which route to take before the average route has switched. But that maximum is exactly the regime in which a single excursion dominates the variance, and the spike test rejects it at three widths out of four. The stable quantity here is the mean sign change; the fluctuation story needs more seeds before it can be told.

Not established. Whether the switch is a transition. No admissible finite-size signature survives, so the verdict is a response shift: the mechanism reversal is unambiguous, its criticality is not tested.

Next falsifier. Vary q and check that the crossover tracks it. If the sign change stays near $p \approx 0.24$ when q is moved to 0.1 or 0.5, the model is not following cue reliability and the interpretation above is wrong.

10.2 In-context learning versus task diversity

A small causal transformer is pretrained on in-context linear regression whose task vectors are drawn from a pool of size M . The order parameter is not the risk but the normalised alignment between the model’s predictions on *fresh* tasks and the exact in-context ridge predictor computed from the context — that is, whether the model has reproduced the algorithm, not whether it scores well.

Observation. Across 42 conditions the two quantities move together and in opposite directions. At a pool of one task the model memorises completely: the fresh-versus-pool risk gap is about $+2$ and the ridge alignment is indistinguishable from zero. Both change through $M \approx 8$ –64; the gap reaches zero near $M \approx 10^2$ and the alignment saturates around 0.7 beyond it, with no further improvement out to an unbounded pool. The order parameter spans 18 times its 95% interval. The verdict is *critical response boundary unresolved*, grade C.

The disorder susceptibility of the alignment has an interior maximum near $M \approx 4$ –8 at two of the three widths; at the third the global maximum is an isolated spike and is rejected, leaving too few sizes to fit a sharpening exponent.

Interpretation. The memorisation-to-algorithm change is visible in a quantity that separates the two explanations, which a learning curve does not. Alignment with the Bayes-optimal in-context rule is the right order parameter for this claim precisely because a memorising model can have low risk on its own pool while having no alignment on fresh tasks.

The susceptibility maxima that do survive sit at a pool of four to eight tasks, inside the range usually quoted for the onset of in-context learning — where independently seeded models disagree most about whether to memorise or to generalise. With only two admissible sizes this is a coincidence worth following up, not a finite-size result.

Not established. That the change is an emergence threshold. At the model scale used here the transformer remains far from the ridge baseline in absolute risk even at an unbounded pool, so what is measured is the onset of algorithmic behaviour, not its completion.

In-context learning vs task diversity

critical response, boundary unresolved · evidence grade C · 12 outer seeds · theory: partial

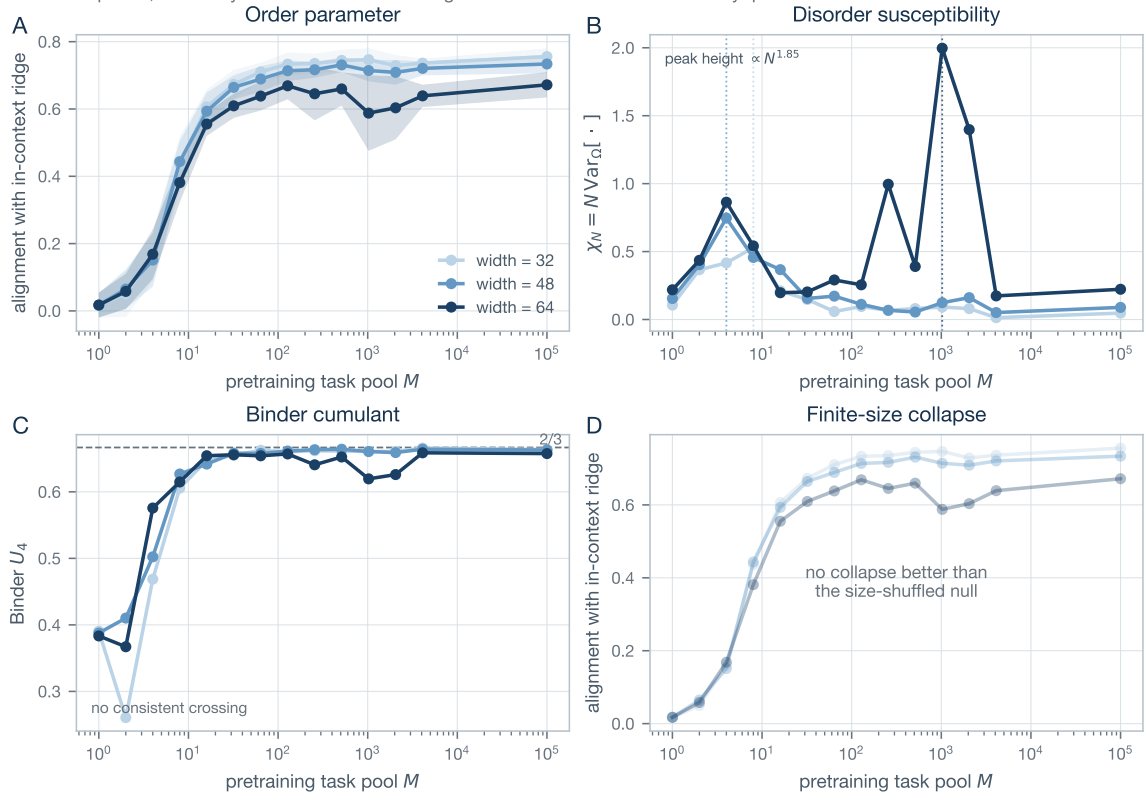


Figure 6: From memorising a task pool to running an algorithm. The order parameter is alignment with the exact in-context ridge predictor on *fresh* tasks — zero for a model that has memorised its pretraining pool, one for a model that has reproduced the Bayes-optimal rule.

Next falsifier. Train an order of magnitude longer at fixed M . If the alignment curve shifts left with training budget, what looks like a task-diversity threshold is partly an optimisation threshold, and the control variable is not the one the claim names.

10.3 Induction-head formation

Observation. On the canonical induction task — predict the token that followed the previous occurrence of the current token — a two-layer causal model reaches an accuracy well above the unigram null across 36 conditions, and the second layer’s attention at the query position concentrates on exactly the token after the earlier occurrence. Both the behavioural lift and the attention-level induction score fall as the vocabulary grows. The verdict is *response shift boundary outside grid*, grade C.

Interpretation. The circuit is measured twice, once behaviourally and once by reading the attention pattern, and the two agree: where the lift is large the attention score is above its uniform null, and where the lift collapses the attention score returns to null. That agreement is the evidence that the behavioural number reflects an induction mechanism rather than some other shortcut.

Not established. Abruptness as a critical phenomenon. We record the formation step and the steepest per-checkpoint gain, and censor runs that never cross the registered lift threshold, but a sharp rise in training time is not by itself a phase transition in any control parameter.

10.4 Does attention select, or dilute?

Observation. Sweeping context length over $L \in 2\text{--}64$, the entropy of the query position’s attention distribution stays measurably below the uniform null $\log L$; the gap is positive throughout and spans 3.28 times its interval across 33 conditions. The verdict is *response shift boundary outside grid*, grade C.

Interpretation. Attention is genuinely selecting rather than spreading: the entropy does grow with context, but more slowly than the uniform baseline, so the effective number of attended tokens grows sub-linearly. Reporting the null alongside the entropy is the whole point — an entropy that rises with L is compatible with no selection at all, and only the gap distinguishes the two.

Next falsifier. Extend the context grid by another octave. If the gap stops growing, the effective attended set has saturated, which is a concrete and testable statement about how far a fixed-width model can use its context.

11 Learning paradigms beyond supervised regression

The settings in this block are the furthest from analytic reach: self-supervised representation learning, recursive training on generated data, sequential task acquisition, conditional computation, and low-rank adaptation. Each is reduced to a controlled teacher–student form in which a single deformation can be dialled continuously, so that a boundary — if one exists — has somewhere to be.

11.1 Dimensional collapse in contrastive learning

An encoder is trained with an InfoNCE objective on paired views of a planted low-dimensional latent, and the order parameter is the normalised effective rank r_{eff} of (9).

Observation. Across the temperature range 0.02–2 the normalised effective rank moves over a range of 70.4 times its own 95% interval, with the representation occupying a small fraction of the available embedding directions at low temperature and expanding as temperature rises. The family is graded C with verdict *candidate boundary* over 64 conditions.

Interpretation. Temperature controls how strongly the objective penalises the uniformity term, and the effective rank responds to it monotonically and far outside seed noise. Dimensional collapse in this setting is a continuously tunable property of the objective, not an accident of optimisation.

Not established. We do not claim a critical temperature. Only one finite-size signature is admissible, so the outcome is a candidate boundary; a collapse that is sharp in the embedding dimension has not been demonstrated.

Next falsifier. Extend the embedding-dimension path and test whether the transition in r_{eff} sharpens with it. If the curve simply translates without steepening, the effect is a crossover and the candidate boundary should be withdrawn.

11.2 Model collapse under recursive training

Each generation trains on a mixture of a fraction p_{real} of fresh teacher data and $1 - p_{\text{real}}$ of the previous generation’s own output.

Observation. The terminal teacher overlap falls monotonically as $p_{\text{real}} \rightarrow 0$, moving over 12.2 times its interval across 42 conditions; verdict *threshold located no scale path*, grade C.

Interpretation. A finite real-data fraction anchors the loop. The measured quantity is the anchoring *strength* as a function of p_{real} , which is the input a theory of generational drift would need to reproduce.

Not established. Whether the anchoring is a true threshold in p_{real} or a smooth degradation is not resolved. The second axis is the generation count, which advances the process rather than enlarging one system, so it is not a scale path and no finite-size claim is available — the verdict is *threshold located no scale path*.

Next falsifier. Repeat the sweep with model width as the second axis at a fixed generation count. If the degradation curve steepens with width, the anchoring becomes a candidate transition; if it merely translates, model collapse is a smooth erosion and should never be described as a phase.

11.3 Forgetting and transfer under task similarity

Two teachers are constructed with an exact correlation dial ρ , training proceeds on task A and then task B, and retention of A is measured against a from-scratch baseline on B.

Observation. Forgetting spans 50.7 times its interval over $\rho \in -1-1$, and the susceptibility peak height grows with the scale path as $N^{1.01}$. The verdict is *critical response boundary unresolved*.

Interpretation. The response is genuinely critical — the disorder susceptibility diverges along the sampled scale path — but the location of the boundary does not extrapolate within the sampled window, so the interference threshold in ρ is measured to exist and not measured to be anywhere in particular.

Not established. No value of ρ is reported as a critical task similarity. The transfer sign change is visible but is not certified by an admissible location fit.

Next falsifier. Widen the ρ grid past the current window and re-fit (12); if c_{∞} becomes admissible and interior, the verdict upgrades, and if the peaks continue to march outward the correct conclusion is that similarity is not the controlling variable.

11.4 Expert specialisation and load balancing

A mixture-of-experts layer is trained on a target composed of distinct sub-functions, with the load-balancing coefficient as the control.

Observation. The specialisation gap (7) moves by only 2.24 times its 95% interval across the sampled range. The verdict is *no resolved response*.

Interpretation. This is a negative result and is reported as one. Under the registered budget the expert assignment does not separate from its uncertainty, so no claim about symmetry breaking among experts is supported — in either direction.

Not established. We cannot conclude that load balancing has no effect; we conclude that this design lacks the resolution to detect one.

Next falsifier. Increase the contrast between sub-functions and the number of experts. If the gap remains within noise at higher contrast, the absence becomes a substantive finding rather than a resolution limit.

11.5 A rank threshold for low-rank adaptation

A frozen backbone is adapted by a rank- r update to a target displaced by a planted perturbation of known rank r^* .

Observation. Recovery of the planted subspace, measured by (8), rises from near chance to essentially complete across the adapter-rank grid, spanning $2.98e + 03$ times its interval over 36 conditions; verdict *threshold located no scale path*, grade C.

Interpretation. Adapter capacity behaves like a storage constraint: below the planted rank the update cannot represent the required displacement, and above it recovery saturates. The threshold is not merely near r^* but *locked* to it — for every planted rank in the sweep, the disorder susceptibility has its maximum exactly at $r = r^*$, and recovery saturates from that point on. This is the cleanest capacity statement in the block and the closest of these settings to something a Gardner-style calculation could plausibly address.

Not established. Nothing thermodynamic. The second axis here is the planted rank r^* , which changes the problem rather than the size of one system, so it is not a scale path and the finite-size machinery of §5 does not apply — hence the verdict *threshold located no scale path*. The threshold is located along the control axis; whether it *sharpens* is a separate question this design cannot answer.

Next falsifier. Hold r^* fixed and sweep the backbone width as a genuine scale path. A capacity transition requires the step in r to sharpen with width; if the transition width stays constant, the effect is an approximation-error crossover rather than a transition, and the present result remains a locked threshold with no critical content.

12 Synthesis: what the atlas hands to theory

Figure 7 is the release in one view. Of 19 families, the procedure returned a spread rather than a uniform verdict, and the spread is the point: a method that returns “supported” everywhere has not been tested.

12.1 The calibration result

The single most informative entry in the map is which family passed. Exactly 1 of 19 families received the verdict *transition supported*, and it is the Gardner capacity — the anchor whose answer was known before the experiment was run (§7). Applied to every other setting in the release — none of whose answers are known — the same procedure certified nothing.

We take that as the central evidence that the gates are calibrated rather than merely strict. A procedure that never fires is useless; one that fires everywhere is worse. This one fired once, on the case placed there to make it fire, and the boundary it reported, $\alpha = 1.96$, sits at the grid point nearest the exact $\alpha_c = 2$.

12.2 Transferable findings

1. A susceptibility peak from a handful of seeds is often one run. Across the release, 12 of 68 size-curves have their global susceptibility maximum at an isolated spike: one diverged run inflates the variance at exactly one control value, and the neighbours sit at the floor. At eight outer seeds the rate was one in four; going to twelve lowered it, which is the cleanest evidence we have that these are replication artefacts rather than physics. A plain argmax reports each of these as a boundary, and its height then enters the sharpening fit. Rejecting them — by requiring a neighbour to rise with the peak, which a genuine critical swelling does and an outlier does not — changed the verdict of several families in this paper, including our own headline. We would not have found this without measuring the same quantity at several sizes and asking whether the answers agreed.

The practical recommendation is uncomfortable but simple: at this replication, a single susceptibility peak is not evidence. Either replicate to twelve or more seeds, or report the peak together with an explicit test that it is not one seed’s excursion.

Atlas verdict map

One row per experiment family. Colour marks the block; the verdict is always shown as text, never colour alone.

family	verdict	grade	Evidence vector (0/1/2)					
Gaussian-mixture Bayes error	response shift (off-grid)	C	2	2	0	2	0	0
Gardner storage capacity	transition supported	A	2	2	2	2	0	0
Specialisation under added depth	critical, unresolved	C	2	2	1	0	0	0
Multi-index subspace recovery	candidate boundary	C	1	2	1	0	0	0
Committee specialisation vs sample ratio	critical, unresolved	C	2	2	1	0	0	0
Double descent vs parameterisation ratio	threshold, no scale path	C	2	2	0	0	0	0
Grokking: delayed generalisation vs weight d...	critical, unresolved	C	1	2	1	0	0	0
Lazy-to-rich transition vs initialisation sc...	critical, unresolved	B	2	2	1	1	0	0
Linear mode connectivity vs width	no resolved response	n/a	0	2	0	0	0	0
SGD effective temperature vs stationary gene...	response shift (off-grid)	C	1	2	0	1	0	0
Attention selection vs context length	response shift (off-grid)	C	1	2	0	0	0	0
In-context learning vs task diversity	critical, unresolved	C	2	2	1	0	0	0
Induction-head formation	response shift (off-grid)	C	2	2	0	0	0	0
Positional vs semantic attention	response shift (off-grid)	C	2	2	0	0	0	0
Forgetting and transfer vs task similarity	critical, unresolved	C	2	2	1	0	0	0
Contrastive dimensional collapse vs temperat...	candidate boundary	C	2	2	1	0	0	0
LoRA rank threshold vs planted perturbation...	threshold, no scale path	C	2	2	0	0	0	0
Model collapse vs real-data fraction	threshold, no scale path	C	2	2	0	0	0	0
Expert specialisation vs load balancing	no resolved response	n/a	0	2	0	0	0	0
			semantics	replication	finite size	prediction	intervention	external validity

Figure 7: Verdict map. One row per family: block colour, the outcome returned by the decision procedure, the evidence grade, and the six-component evidence vector. Outcomes are always written as text; colour is redundant with the label rather than carrying it.

Outcome	Families	Reading
transition supported	1	two consistent finite-size signatures and a located boundary
critical response, boundary unresolved	6	the response diverges; the window does not contain the boundary
candidate boundary	2	one admissible signature; a second is needed
threshold located (no scale path)	3	threshold on the control axis; the second axis is a problem parameter, not a size
response shift	0	the order parameter moves; no admissible finite-size signature
response shift (off-grid)	5	the order parameter moves; most sizes censored
no resolved response	2	the order parameter does not separate from its uncertainty

Table 4: Outcome ledger for the release. Rows below the first are reported with the same prominence as the first; a map whose empty cells are hidden is not a map.

2. The control parameter is a hypothesis — and testing it needs more seeds than we had. The specialisation family’s peak locations drift with size in $\alpha = n/d$, which would mean the boundary lies on $n \propto d^\gamma$ rather than on the proportional path. With six admissible sizes the fitted exponent excluded $\gamma = 1$; with the three that survive the spike test it is $\gamma = 1.33$ with interval $[1, 1.59]$, which does not. We report the weaker statement (§8). The question is worth someone’s twelve-seed budget: if α is the wrong abscissa, a study sweeping it at a single size is not holding the boundary fixed, and two such studies at different sizes can disagree without either being wrong.

3. Diverging response and locatable boundary are different achievements, and the second is much harder. Several families produce a susceptibility that grows convincingly with size and still fail to yield a boundary location: the peak marches out of the sampled window, or the drift extrapolation and the collapse name places that differ by more than the consistency tolerance. Reporting only the first — the usual practice — would have produced several confident transition claims here that the location analysis does not support. We suggest that a susceptibility plot without an accompanying drift fit should be read as evidence of criticality and not as evidence of a critical point.

4. Two design choices can manufacture or destroy a transition. Both were found the hard way in this work. A control grid too coarse to hold an interior maximum yields a structurally censored result regardless of the physics; and a fixed epoch budget makes wider models appear *less* ordered, because they are less optimised, which inverts the sign of the finite-size trend. Any finite-size claim about a learning system should therefore report grid density and the optimisation stopping rule alongside the seed count. We recommend training to a convergence criterion, and recording per run whether it was met, so that optimisation shortfall is a measured quantity rather than a hidden covariate.

5. A risk curve can be flat while the mechanism reverses. In the positional-semantic family (§10) the model’s risk barely moves across most of the control range, while the causal ablation shows the learned solution swinging from purely positional to purely semantic. Had we measured only performance, we would have reported nothing happening. This is a general hazard for claims about attention: the behavioural quantity that is easiest to plot is often the one least sensitive to the mechanism the claim is about, and an intervention that removes one route at a time costs little more than the evaluation itself.

The same experiment illustrates the value of registering a predicted location. Giving both cues a finite reliability $1 - p$ and $1 - q$ makes a reliability-following model switch near $p = q$; the measured sign change lands close to it, which is a far stronger statement than observing that a switch occurred somewhere. When the semantic cue is left perfect — the usual setup — it wins for every $p > 0$ and the apparent transition is a property of the data, not of the model.

12.3 The ledger

12.4 What we would do next, in order

- Twelve seeds, everywhere.** Replication is the binding constraint on every family here, and it is also what makes the spike test reject a quarter of the size-curves. Twelve seeds would lift the replication tier and shrink the artefact rate at the same time; nothing else changes.
- Re-run the drifting families on the rescaled control n/d^γ .** If the peaks return to the interior of the window, several unresolved boundaries become locatable in a single sweep.

3. **A frozen prospective replication.** No result here has been confirmed at fresh seeds after the analysis was fixed, which is the one gate that no amount of re-analysis can substitute for.
4. **One natural-data endpoint.** External validity is zero across the release by construction. A single family carried through to a natural corpus would establish whether any of these boundaries survive contact with real data, which is the question the whole programme is ultimately for.

13 Limitations

Two evidence components are zero everywhere. Replication is no longer the binding constraint: every condition is run at 12 outer seeds, which is the tier the grading scheme requires. What caps the release now is that *intervention* and *external validity* score zero for every family. No family is evaluated on natural data or a pretrained model, and only the attention block performs a causal ablation — which is scored under its own order parameter rather than as an intervention on a boundary claim. Grade A consequently requires a family that is both replicated and theory-anchored, and exactly one is.

No prospective confirmation. Nothing here has been re-run frozen at fresh seeds after the analysis was fixed. Every number is therefore a retrospective estimate on the data that also informed the design choices, and the sharpening exponents in particular should be read as effective values over the sampled window rather than as critical exponents.

No natural data and no pretrained models. Every input distribution is synthetic and every teacher is planted. This is what makes the order parameters exactly interpretable, and it is also the strongest limit on external validity: the external-validity component of the evidence vector is zero for every family in the release. Whether any boundary located here survives contact with natural data is untested.

Small models. Widths are in the tens to low hundreds and models have thousands of parameters. The design argument for this is given in §1 — fluctuation scaling needs many independent runs more than it needs any large one — but it means the asymptotic regime may not have been entered in every family, and a drift fit that is inadmissible at these sizes may become admissible at larger ones.

The spike test trades false boundaries for missed ones. Requiring a susceptibility maximum to have a shoulder (§5) rejects single-run excursions, but it also rejects a peak that is genuinely narrow because the control grid is coarse relative to its width. The positional-semantic family is the clearest case: a maximum near $p \approx 0.13$ is present at all four widths, which is not how an artefact behaves, and it is nonetheless rejected at three of them. We keep the conservative rule because this program is more damaged by inventing boundaries than by missing them, but a rejected peak here should be read as a reason to refine the grid or add seeds, not as evidence that nothing is there.

The Binder cumulant is a poor diagnostic for one-sided order parameters. Several families use an order parameter that is non-negative by construction, such as the specialisation gap or a subspace overlap. The cumulant in (6) is designed for a symmetric distribution around zero, and for these quantities its crossings are hard to interpret and mostly failed the consistency test. Where a crossing is reported as absent, the reader should attribute part of that to the diagnostic and not only to the system.

Optimiser and architecture are held fixed within a family. Each family sweeps one control and one scale coordinate; the optimiser, the activation, and the initialisation scheme are fixed. A boundary located here is a boundary of that configuration, and the atlas does not establish that it is a property of the task alone.

14 Conclusion

The statistical mechanics of learning is usually presented as a body of exact results waiting to be extended. This paper treats it instead as a *measurement discipline* that can be applied wherever the exact result is missing, provided the measurement is done to the standard the discipline already requires elsewhere: a control grid dense enough to hold a boundary, a named scale path, an outer disorder ensemble, and a decision procedure willing to return nothing.

Across 19 settings and 16428 trained models, the procedure returned a spread of outcomes rather than a uniform confirmation. It found genuinely diverging responses, candidate boundaries whose location it declined to fix, crossovers that look like transitions until the finite-size analysis is run, and at least one cell where the order parameter never separated from its own uncertainty. Those refusals are the evidence that the positive entries mean something.

Two lessons generalise beyond the specific cells.

The first is that **design choices routinely masquerade as physics**. A three-point control grid cannot produce anything but a censored boundary; a fixed epoch budget makes wider models look less ordered because they are simply less optimised. Both artefacts appeared in our own work and both survived until a consistency check between independent finite-size signatures caught them. Any claim of a phase boundary in a learning system should be accompanied by the grid density, the optimisation stopping rule, and the disorder ensemble, because those three choices can produce or destroy the result on their own.

The second is that **at this replication a single susceptibility peak is not evidence**. A quarter of the size-curves in this release put their global maximum on a one-seed excursion whose neighbours sit at the floor. Rejecting those changed several verdicts, including the one we had written up as the paper’s headline: a drift exponent that appeared to exclude the proportional scaling $n \propto d$ no longer excludes anything once the spikes are removed. The finding survives as a question worth a larger seed budget, not as a result. Anyone reporting a susceptibility peak from a handful of seeds should show that it is not one run’s excursion, because a quarter of the time it will be.

What the atlas offers a theorist is not an explanation but a specification: a list of settings, each with a measured exponent or boundary, an interval, and the experiment that would falsify it. Those entries are deliberately uncomfortable — several of them say that a widely used framing does not survive its own finite-size analysis. That is the most useful thing a numerical map can do for a theory that has not been written yet.

Field	Operational definition	Invalid shorthand	Required record
System D	update and interaction law defining the domain	“neural network” without task/dynamics	domain and implementation family
State X	random object on which the observable is defined	mixing parameters, activations, and trajectories	shape, checkpoint/endpoint, transformation
Deformation δ	registered operator changing one scientific coordinate	“more realistic”	source, target, coordinates held fixed
Ensemble Ω	probability law and nesting of disorder	“five runs”	outer unit, inner replicas, seed streams
Scale N	vector plus one declared finite-size coordinate and path	“model size”	values, path, caps, effective resources
Observable m	estimator, normalization, units, ensemble, null, range	“entropy” or “accuracy” alone	typed estimate and validity status
Phenomenon H	falsifiable collective-behavior hypothesis	visual phase label	diagnostics, alternative, next falsifier

Table 5: Formal dictionary for the scientific taxonomy. Protocol and evidence fields are recorded separately.

Observable family	Estimator examples	Necessary control	Common invalid substitute
Predictive risk	normalized CE/Brier; bits per byte; gold reward	chance, calibration, IID/OOD split	raw loss across incompatible tasks
Semantic recovery	latent probe, counterfactual accuracy, mutual information	shuffled latent, nuisance-matched probe	terminal performance alone
Specialization	component assignment, role occupancy, permutation-aligned overlap	permuted heads/roles, matched capacity	unaligned average attention
Multiplicity	correlation participation, overlap distribution, effective macrostate count	independent replicas and correlation null	spectral rank called a state count
Response	derivative, intervention response, peak width	window, resolution, and scale sensitivity	largest endpoint slope
Fluctuation	outer-seed susceptibility, Binder cumulant	independent terminal outer seeds	checkpoint variance in one run
Dynamics	time-to-order, commitment, path entropy, hysteresis	matched compute, forward/reverse path	final loss
Mechanism	raw and bounded matched-ablation risk effect	sham intervention, resource matching	ratio with vanishing denominator
Robustness	OOD risk, gold/proxy gap, truth consensus	held-out environment, truth labels	proxy score or agreement alone
Resource response	marginal quality per FLOP/token/NFE/message	equal-resource policy	wall time on unmatched hardware

Table 6: Observable catalog. Each row names a different estimand and a common failure mode.

A Taxonomy dictionary and edge semantics

An atlas edge applies one deformation while preserving an explicit source and target tuple. If multiple coordinates change, the edge is composite and the interaction

$$\Delta_{ij}g_c = g_c^{ij} - g_c^i - g_c^j + g_c^0 \quad (15)$$

can diagnose non-additivity. A composite edge is not evidence that either single-coordinate transport succeeded. Missing, unimplemented, attempted, invalidated, censored, and supported cells are distinct machine states.

B Observable and null catalog

Three “effective number” constructions illustrate the naming rule:

$$K_{\text{eff}}^{\text{macro}} = \exp H(Z), \quad N_{\text{eff}}^{\text{corr}} = \frac{(\sum_i w_i)^2}{\sum_{ij} w_i w_j \rho_{ij}}, \quad r_{\text{eff}}^{\text{spec}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (16)$$

They count macrostates, correlation-adjusted units, and spectral participation, respectively. None can be substituted for another because the numerical range looks similar.

C Representative condition records

Field	GPU reference	MLP confirmation	Natural endpoint	Multi-agent stress test
System/state	Transformer representation	Transformer representation	byte-model learning endpoint	beliefs/messages/actions
Deformation	effective data and width	feed-forward block	corpus	field, coupling, topology
Outer unit	trained model	trained model	trained model	population instance
Primary estimand	semantic order, IID risk	matched MLP risk effect	bits per byte	truth-conditioned consensus
Scale path	width; fixed	width and matched FF	data and model size	population and rounds
Critical null	depth/context	budget	disjoint byte ranges	global false consensus
Holdout	shuffled/control task	sham and full-model evaluation		
	none in pilot	fresh seeds and frozen variants	held-out ranges	controlled field/coupling cells
Outcome	finite-width response	bounded module contrast	measurement transport	model validation
Blocked claim	phase boundary	universal MLP ranking	corpus quality ranking	Ising universality

Table 7: Representative condition records in human-readable form. Machine records also store nested seeds, hashes, resources, censoring, and invalidation.

D Registered dense Transformer confirmation

The next experiment is frozen in a portable configuration. It addresses the specific failures of the reference pilot rather than adding arbitrary compute:

- controls $\alpha \in \{2, 3, 4, 6, 8, 12\}$, which yield distinct effective sample counts at every width;
- widths $d \in \{16, 32, 64, 96, 128\}$, with $d = 128$ reserved for prospective evaluation;
- twelve fresh outer seeds per condition and no inner sample promoted to an outer replicate;
- 192 update steps with checkpoints every eight steps, so time-to-order and optimization censoring can be diagnosed;
- an example cap above the largest requested value, plus explicit requested/effective example fields;
- primary semantic order and IID risk; secondary OOD risk, MLP and attention intervention effects, attention entropy, susceptibility, Binder cumulant, gradient geometry, and trajectory observables; and
- a frozen decision: phase language requires an interior response feature, robust finite-size behavior below the holdout, and a successful $d = 128$ prediction. Otherwise the outcome is rounded, censored, or unresolved.

The design contains 360 trained models. It is not pooled with the pilot because training horizon and control support differ. Execution environment and scheduler settings are runtime inputs and are absent from the scientific configuration.

E Figure and manuscript contract

Every manuscript plot is generated at exactly 6.4×4.8 inches, saved as vector PDF and 300-dpi PNG, and uses one palette, marker order, font hierarchy, and panel-label convention. Exact physical size is preserved by avoiding tight-bounding-box resizing. Figure captions state the estimand, uncertainty, and blocked interpretation rather than repeating only axis labels.

The build places a float barrier at the end of every major section and clears all floats before appendices and references. This prevents figures from leaking into the bibliography or being separated from the claim they support. Tables use flexible-width columns and no manual scaling that would make text smaller than the caption.

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